Algebra Monsoon 2026 Exercise : Two presentations of the dihedral grou

Fix an integer $n \geq 1$.

Find an isomorphism between the groups G and H :

$$G = \langle a, c \mid a^2 = e,\ c^n = e,\ aca^{-1} = c^{-1} \rangle$$

and $$H = \langle b, d \mid b^2 = e,\ d^2 = e,\ (bd)^n = e \rangle.$$

Consider the unique morphism of

groups $\varphi : \underbrace{\langle a, c \rangle}_{\text{the free group on } \{a,c\}} \longrightarrow \langle b, d \rangle$

such that $\varphi(a) = b$,

$\varphi(c) = bd$.

We have $\varphi(a^2) = b^2 = e$,

$\varphi(c^n) = (bd)^n = e$, and

$\varphi(aca^{-1}c) = \underline{b \cdot bd \cdot b^{-1}} \underline{bd}$

$= ededd = d^2$

$= e$,

so there is a unique morphism of groups $\overline{\varphi} : G \longrightarrow H$ such that $\overline{\varphi}(a) = b$ and $\overline{\varphi}(c) = bd$.

Consider the unique morphism of groups

$\psi : \underbrace{\langle b, d \rangle}_{\text{the free group on } \{b,d\}} \longrightarrow G$

$\psi(b) = a$ and

such that $\varphi(d) = ac$.

We have $\varphi(b^2) = a^2 = e$,

$$\varphi(d^2) = (ac)^2 = acac$$
$$= aca^{-1}c = e,$$

and $\varphi((bd)^n) = (a.ac)^n = (ec)^n = c^n = e$,

so there is a unique morphism of groups $\overline{\varphi} : H \longrightarrow G$ such that

$\overline{\varphi}(b) = a$ and
$\overline{\varphi}(d) = ac$.

We have

$$\psi(\varphi(a)) = \psi(b) = a$$

and $\psi(\varphi(c)) = \psi(bd) = a.ac$
$= a^2c$
$= c$

So $\psi \circ \varphi = Id_G$.

Similarly,

$$\varphi(\psi(b)) = \varphi(a) = b$$

and $$\varphi(\psi(d)) = \varphi(ac) = b \cdot bd$$
$$= b^2 d$$
$$= d,$$

so $$\varphi \circ \psi = Id_H .$$

It follows that φ is an isomorphism and $\varphi^{-1} = \psi$.

Determining G :

- $C = \langle c \rangle$ is a <u>normal</u> subgroup of G because $a c a^{-1} = c^{-1}$.
- $G/C = \langle a \mid a^2 = e \rangle$ has order 2.

So $$G = C \cup aC$$
$$= \{ e, c, \dots, c^{n-1}, a, ac, \dots, ac^{n-1} \}.$$

$a, a^2, \ldots, a^{n-1}, b, ab, a^2b, \ldots, a^{n-1}b$

is the dihedral group of order $2n$.